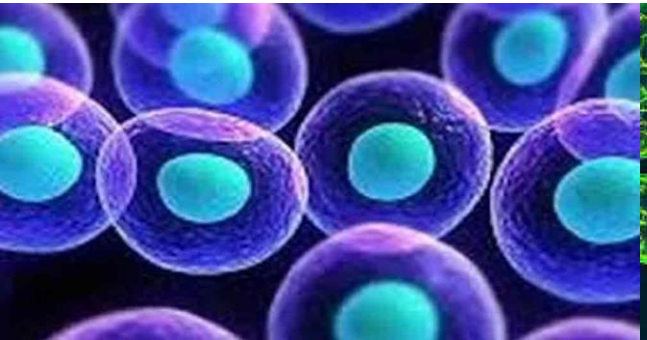
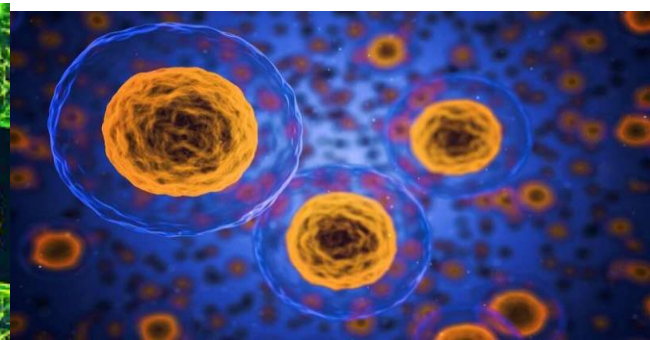
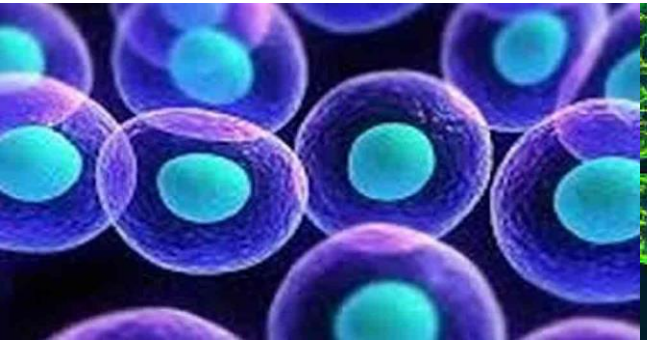
Part 7

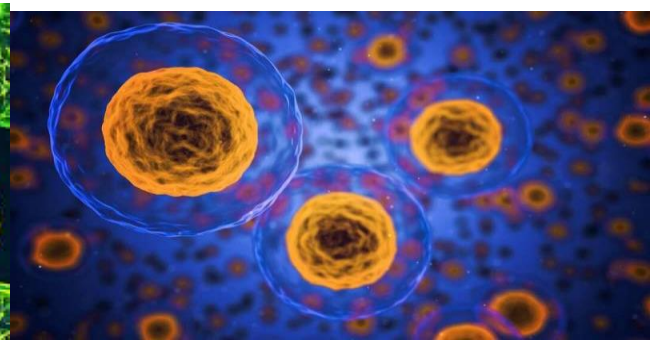
Cell cycle
Cell division





Cell cycle

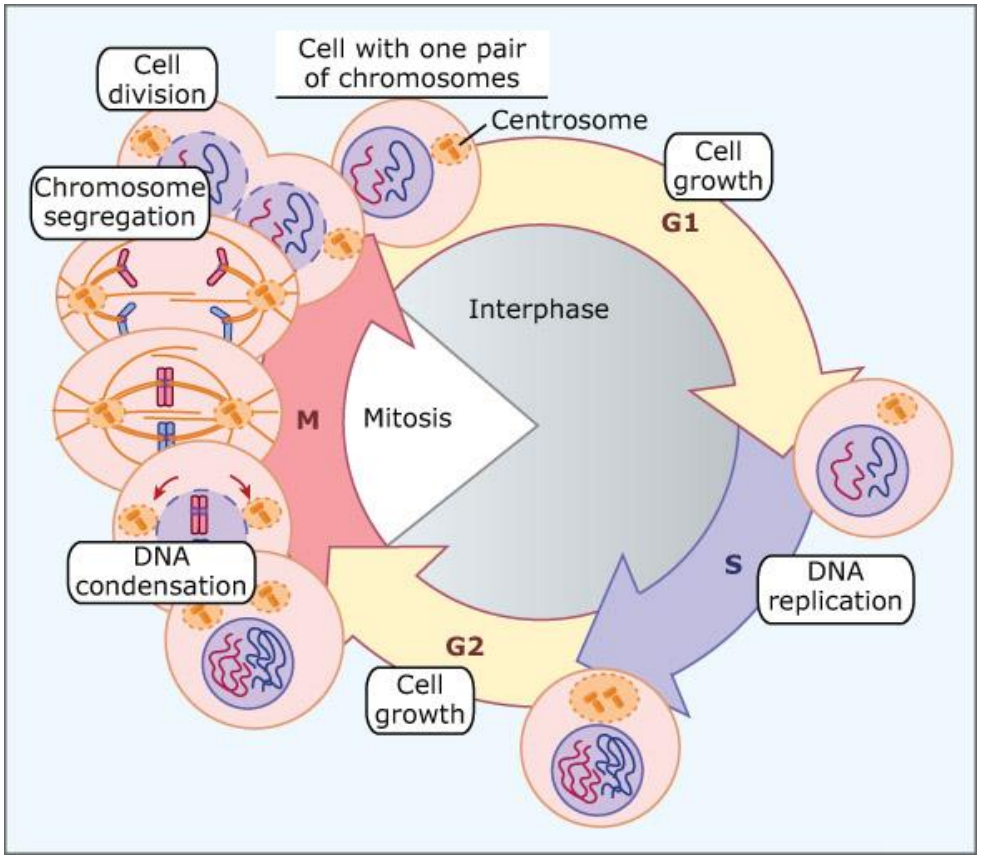




Overview of the cell cycle

- Interphase
  - + G1 phase
  - + S phase
  - + G2 Phase
- Mitosis phase

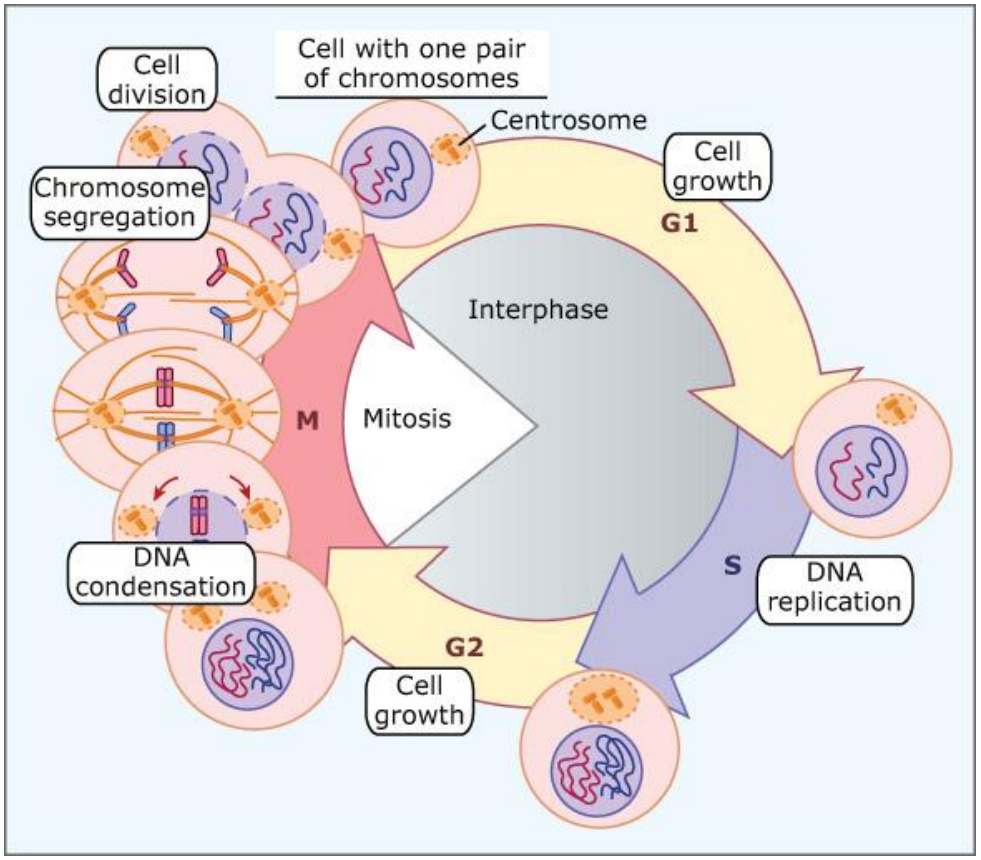
* Meiosis



This cycle lasts about 24 hours. The **DNA replicative phase (S)** and **mitosis (M)** are separated by **Gap phases (G1 and G2)**. The chromosomes, indicated by blue and red lines, are contained within the nuclear envelope until mitosis. Centrosomes serve as organizing centers, or poles, for the mitotic spindle during mitosis and are duplicated during the S phase. Each centrosome contains two rod-shaped centrioles in its center

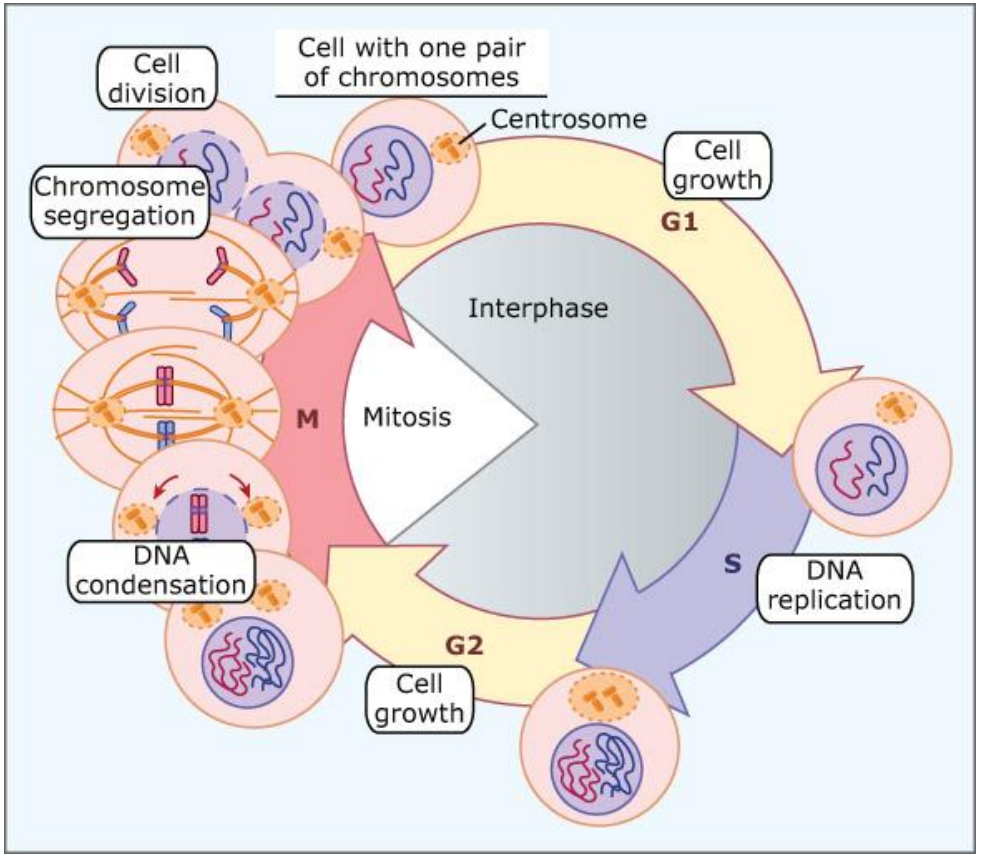
Inter phase: G1 phase

- Recovery from the previous division
- Cell doubles its organelles
- Cell grows in size
- Accumulates raw materials for DNA synthesis (DNA replication)



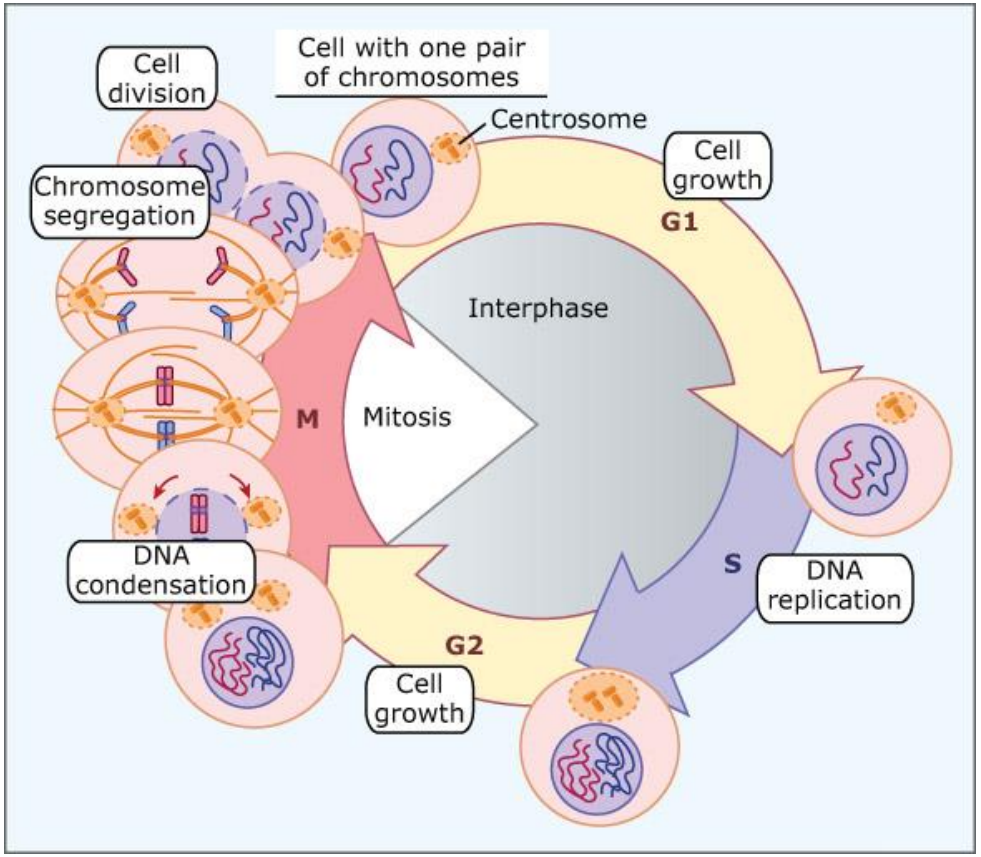
Inter phase: S phase

- DNA replication
- Proteins associated with DNA are synthesized



Inter phase: G2 phase

- Between DNA replication and the onset of mitosis
- Cell synthesizes proteins necessary for division



Cell cycles: Mitosis phase

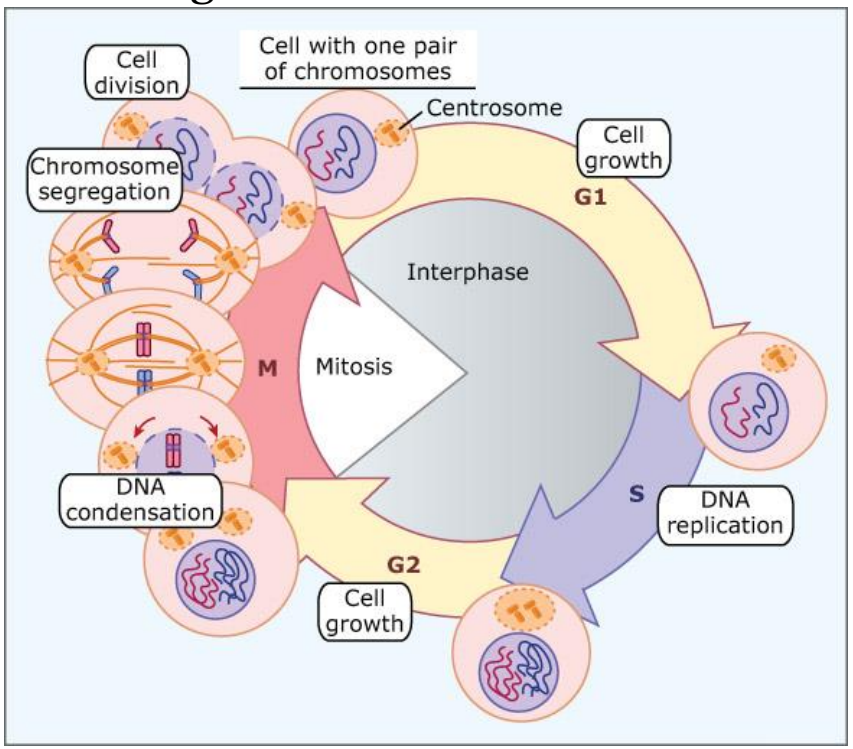
Includes:

Karyokinesis + Nuclear division

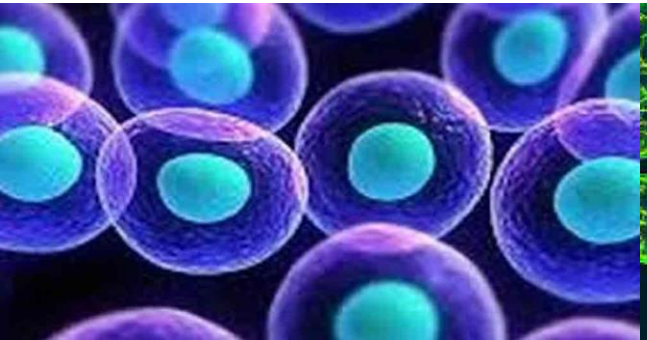
+ Daughter chromosomes distributed to two daughter nuclei

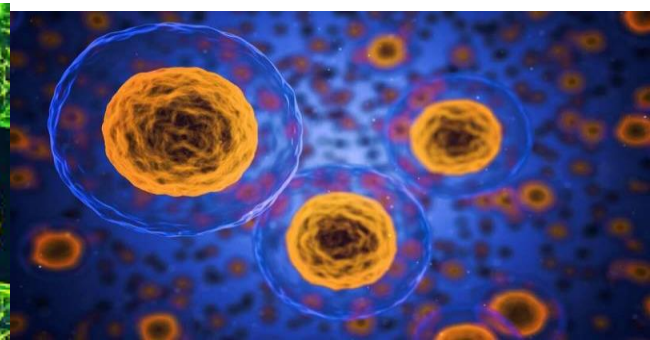
Cytokinesis + Cytoplasm division

+ Results in two genetically identical daughter cells



Intracellular control of the Cell cycle





Intracellular control of the cell cycle

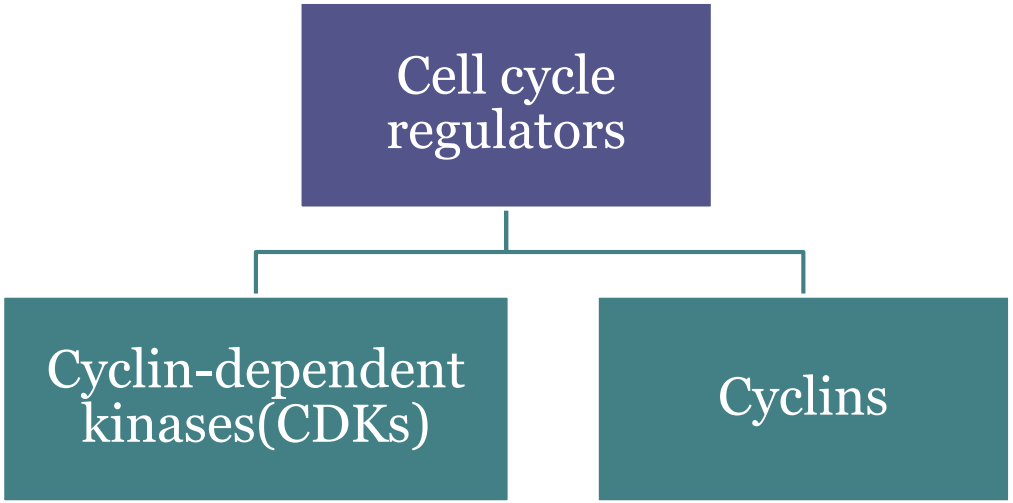
The cell cycle is controlled by regulator molecules that either:

+ Promote the process (positive)

+ Stop it from progressing (negative)

Positive

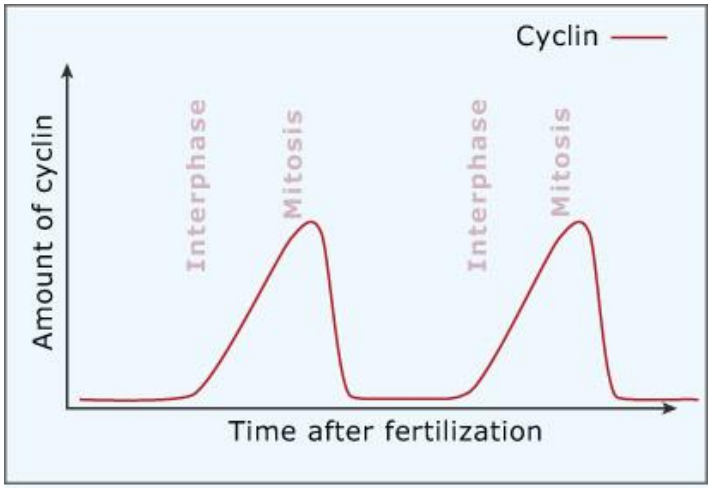
The factors that a cell considers when deciding whether or not to move forward through the cell cycle



Cyclins: the regulatory subunits of the protein kinases that control the cell cycle

Cyclin-dependent kinases: The catalytic subunits of the protein kinases
Must be associated with a cyclin in order to be activated

Cyclin



Scientists discovered the first cyclins when they noted that high cyclin levels were associated with the onset of mitosis in embryos.

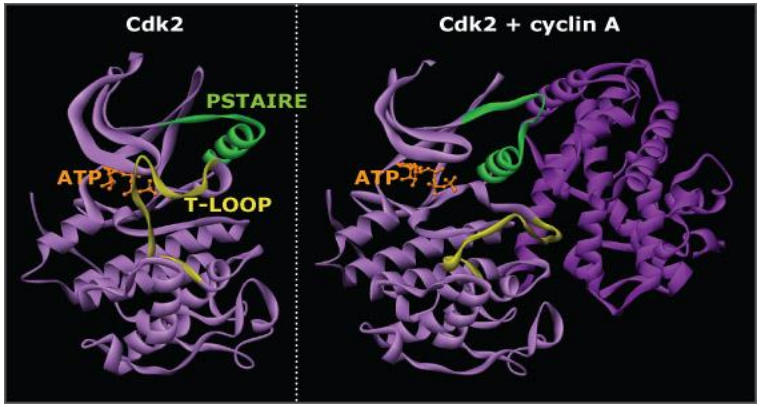
Based on the phase of the cell cycle during which they are present, they form three classes:

- G1 cyclins are responsible for moving the cell cycle through G1 and toward S phase,
- S-phase cyclins are required to initiate and maintain DNA replication,
-M-phase or mitotic cyclins initiate mitosis.

Cyclin-dependent kinases (CDKs)

- CDKs are **kinases**, enzymes that phosphorylate (attach phosphate groups to) specific target proteins

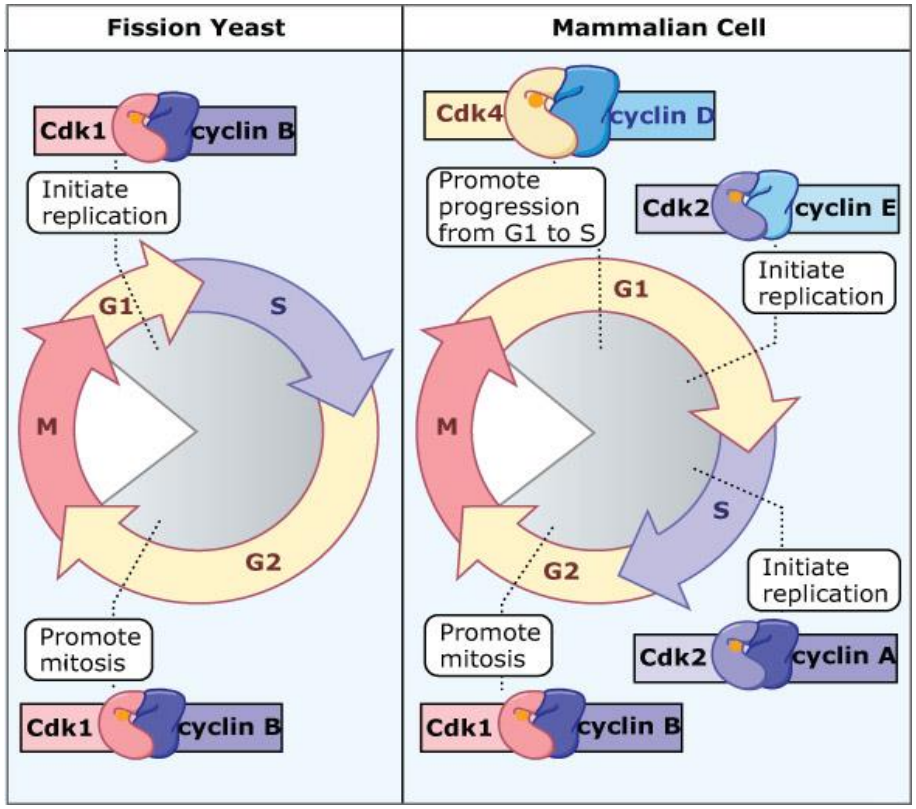
- Active only when complexed with a cyclin protein



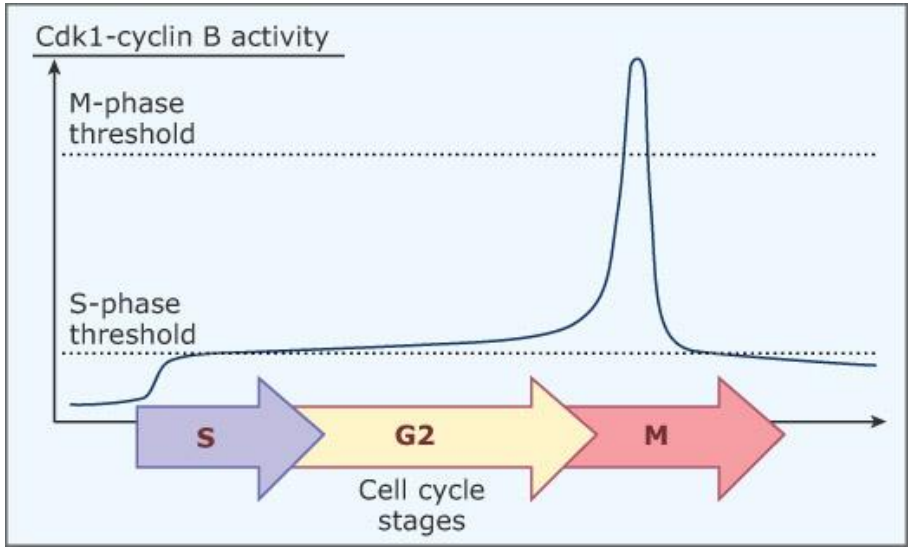
Cyclin binding to CDK causes a conformational change in the CDK. The T-loop changes position, allowing substrate access

- Cyclins and CDKs are very evolutionarily conserved

- Yeast has just one CDKs, humans and other mammals have multiple CDKs that are used at different stages of the cell cycle



A single CDK-cyclin complex (Cdk1–cyclin B) can drive different cell cycle transitions in fission yeast (left), whereas different CDK-cyclin complexes accomplish these tasks in mammalian cells (right)

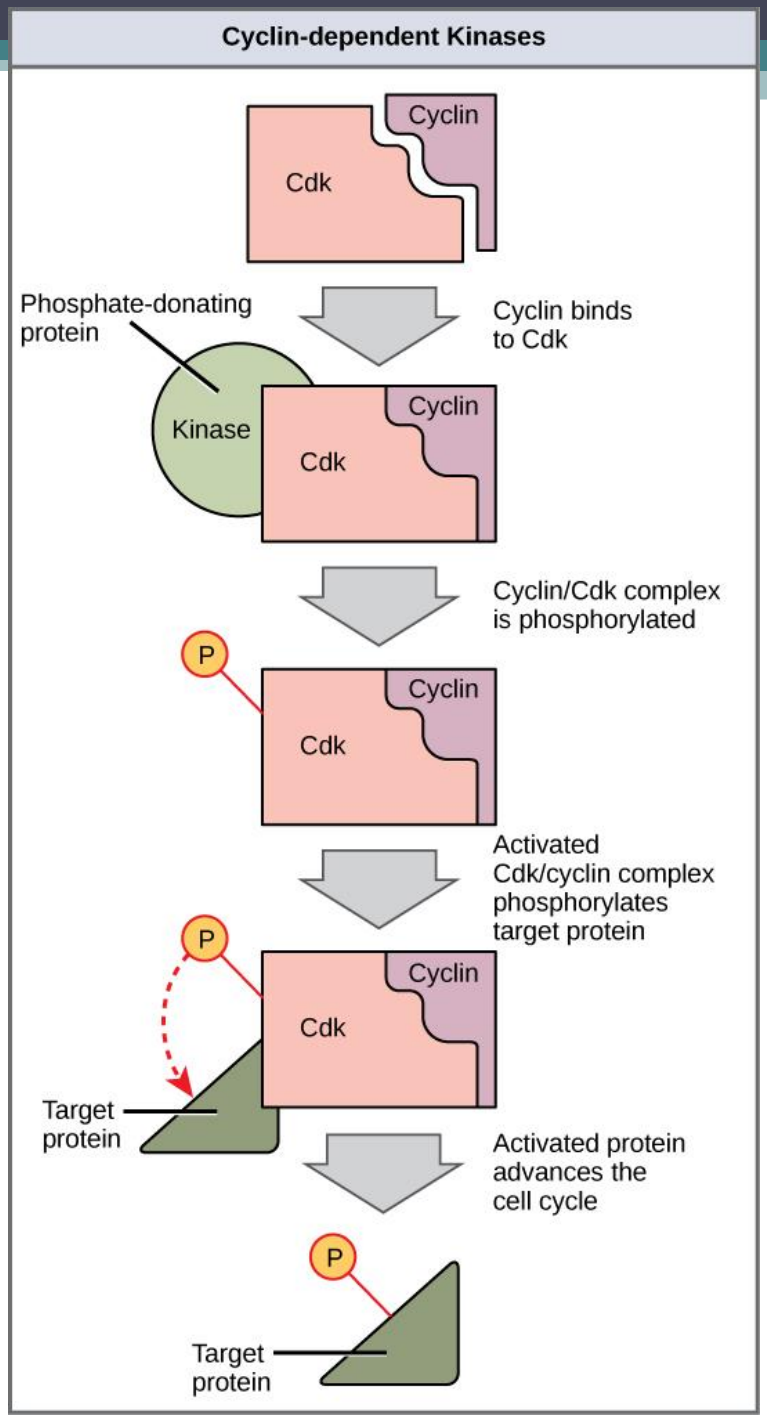


The graph illustrates that different thresholds of a single Cdk1 activity stimulates different events during a fission yeast cell cycle.

Different levels of CDK–cyclin activity are required for the two transitions:

a lower level promoting S phase

a higher level promoting mitosis


Cyclin-dependent Kinases
Cyclin
Cdk
Phosphate-donating protein
Cyclin binds to Cdk
Kinase
Cyclin
Cdk
Cyclin/Cdk complex is phosphorylated
P
Cyclin
Cdk
Activated Cdk/cyclin complex phosphorylates target protein
P
Cyclin
Cdk
Target protein
Activated protein advances the cell cycle
P
Target protein

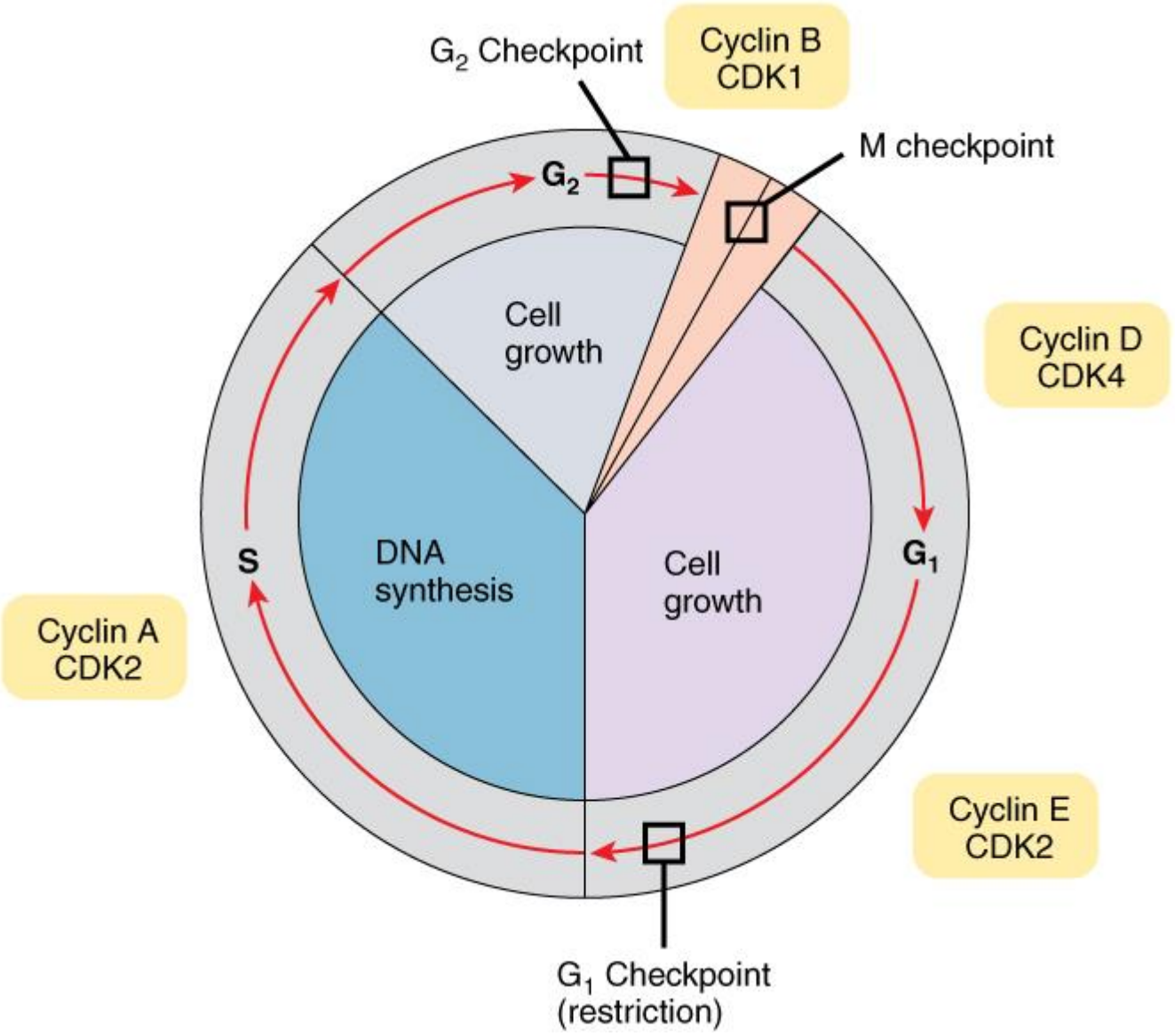
Negative

- Tumor suppressor genes

Tumor suppressor gene code for a signaling protein in an inhibitory pathway. If tumor suppressor fen mutates, the end result can be activate cell division

+ Retinoblastoma protein (Rb): Prevents cell moving into the S phase by binding to a transcription factor. When Rb is phosphorylated it can not bind so the cell can move into the S phase

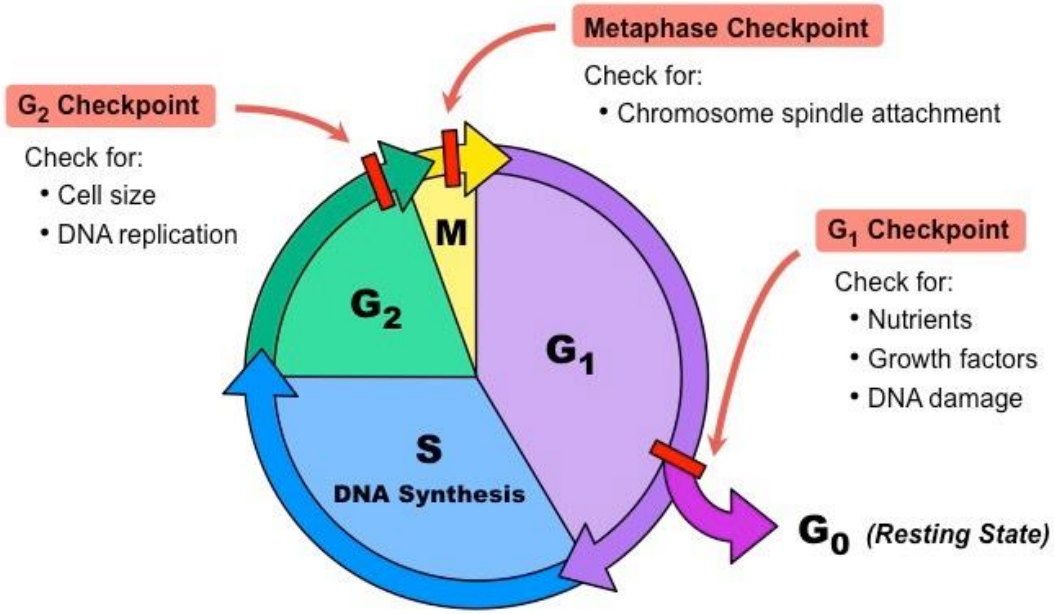
+ p53: Prevents damaged from dividing



How do they work?

Check Point

Together, all the checkpoints ensure that each daughter cell receives a complete and intact copy of the genome



Each step in the cell cycle is monitored to prevent the cycle from proceeding when there is trouble.

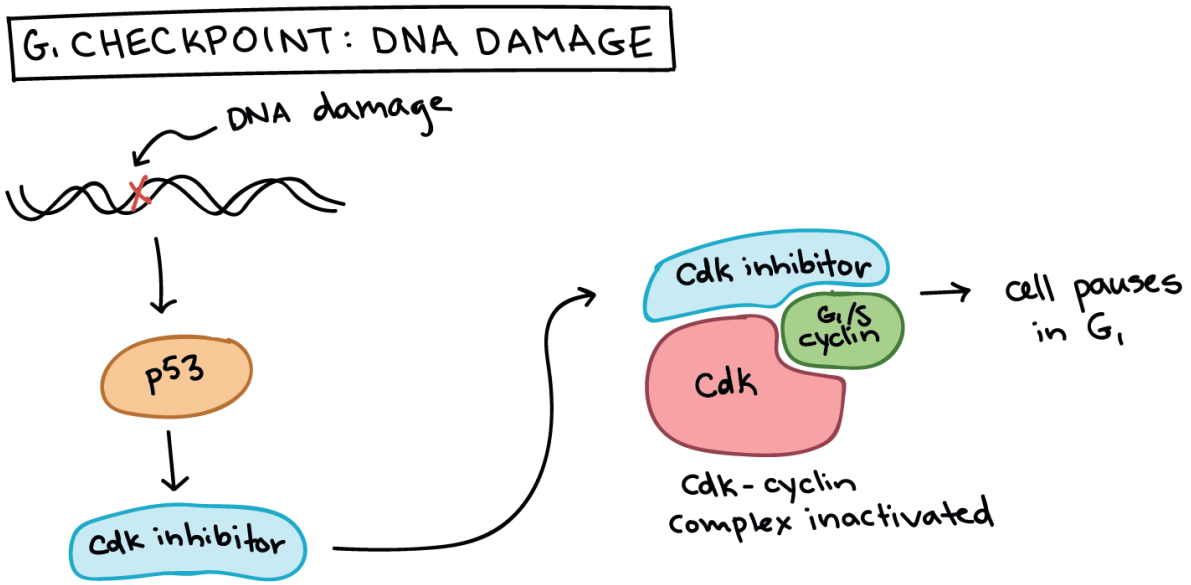
Check Point

DNA damage→ try to fix→ if it is impossible→ preventing the cell division

P53: -Stop the cell cycle at G1 state

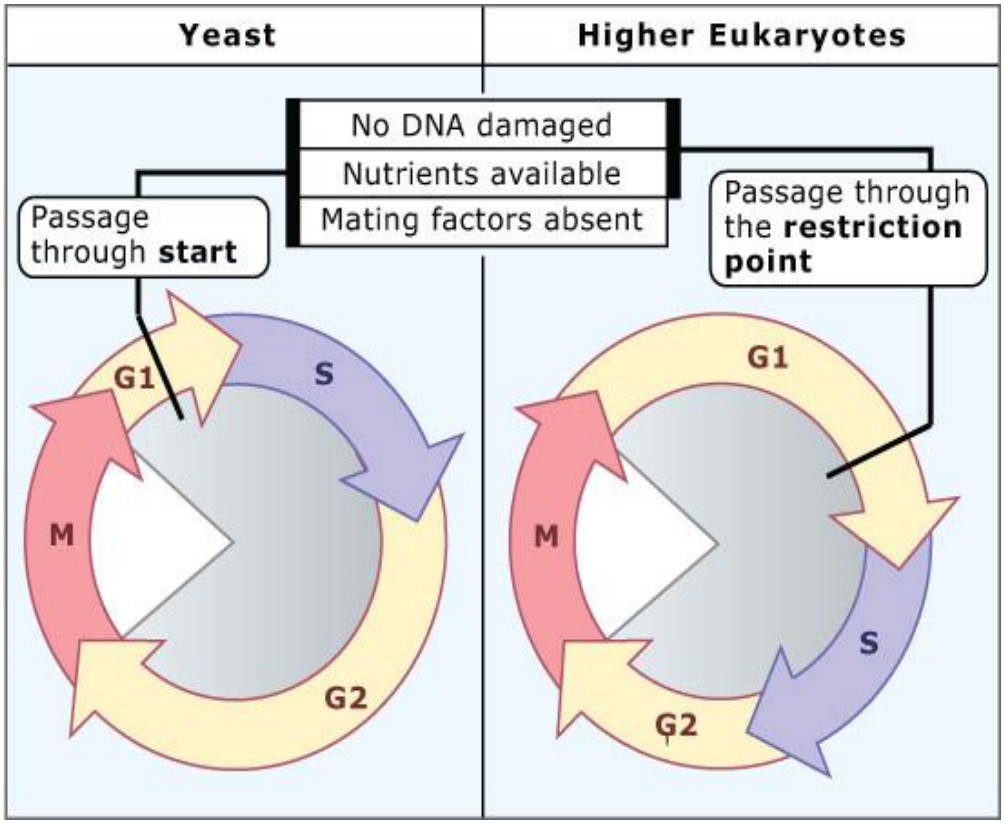
-Activate the DNA repair enzymes

-Triggering programmed cell death



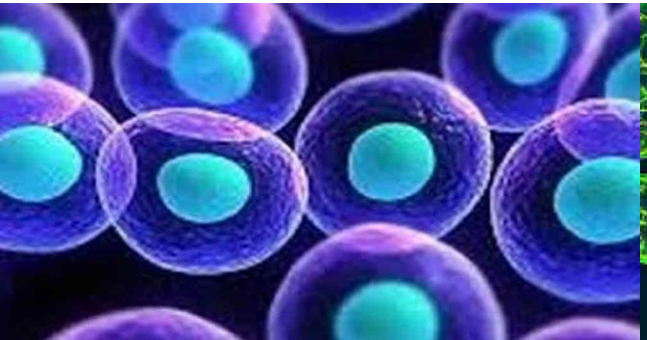
Other function of p53: Forces cell into G0 resting stage

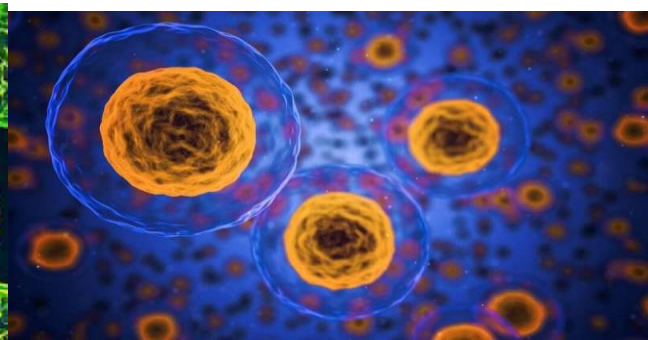
Cells may withdraw from the cell cycle → Go state



Start in yeast and the **restriction point** in multicellular eukaryotes: commitment point at which cells commit to beginning (and completing) a cycle of replication

Cell division



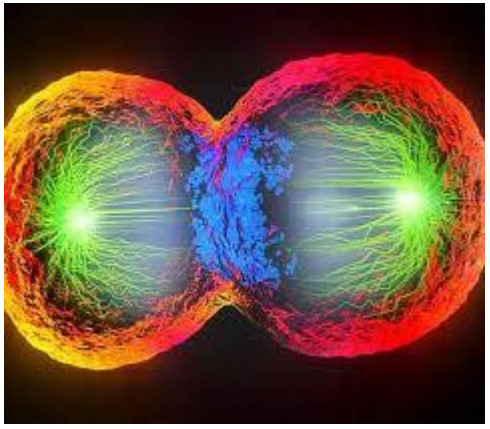


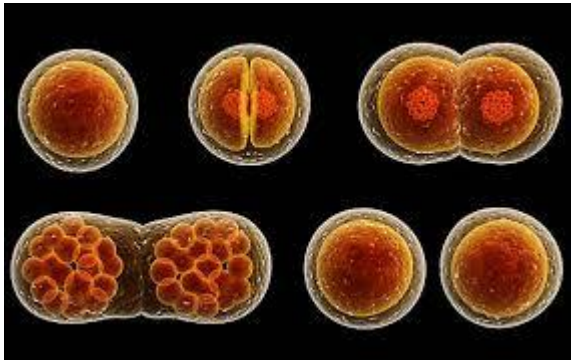
The key roles of cell division

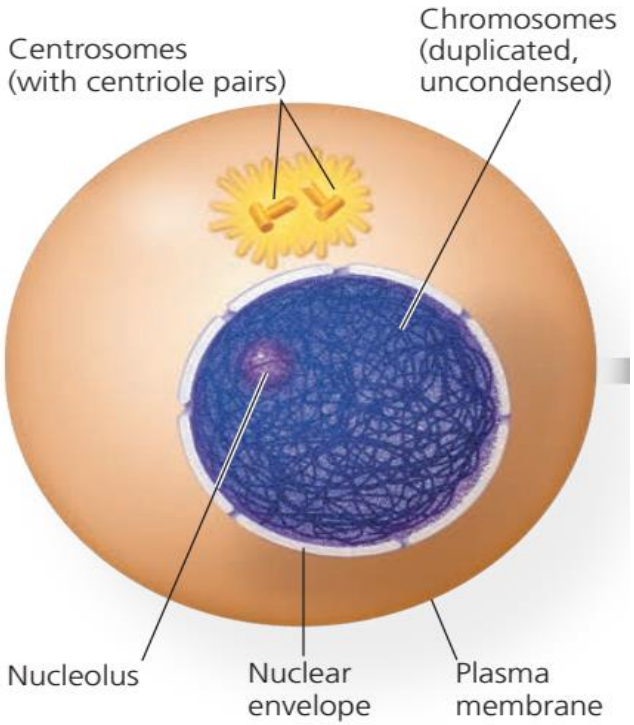
The ability of organisms to reproduce best distinguishes living thing from non-living matter

The continuity of life is based upon the reproduction of cells, or cell division

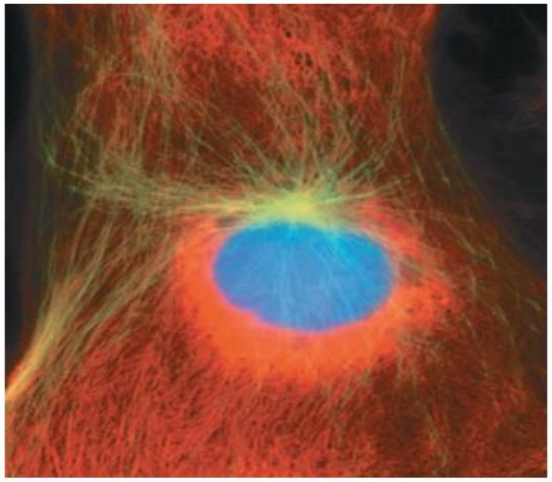
The cell division is integral part of cell cycle



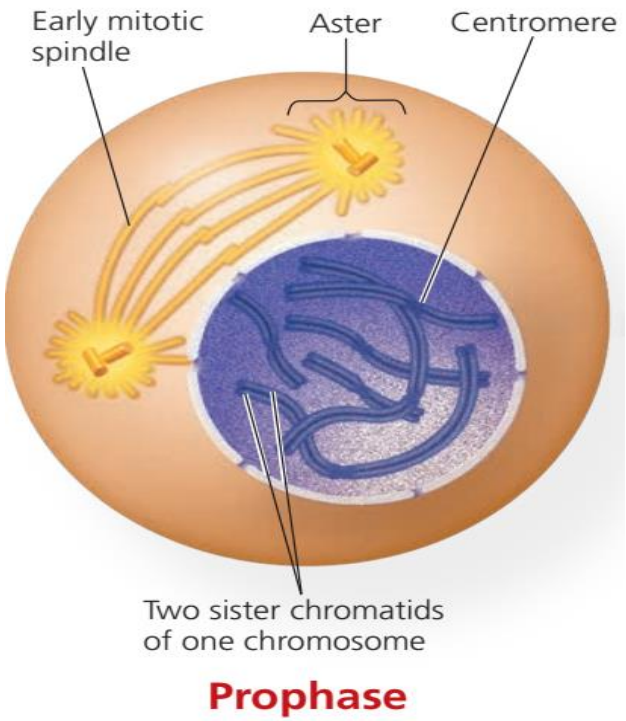




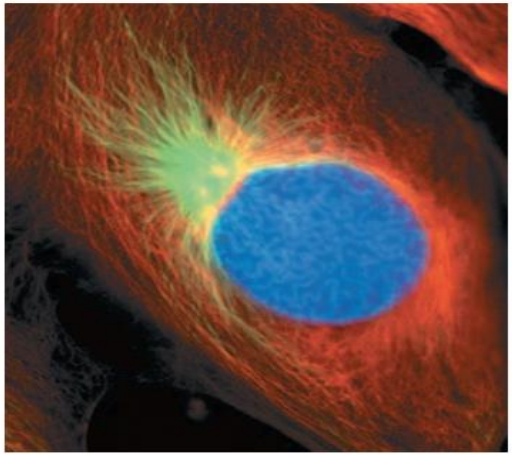
G_2 of Interphase



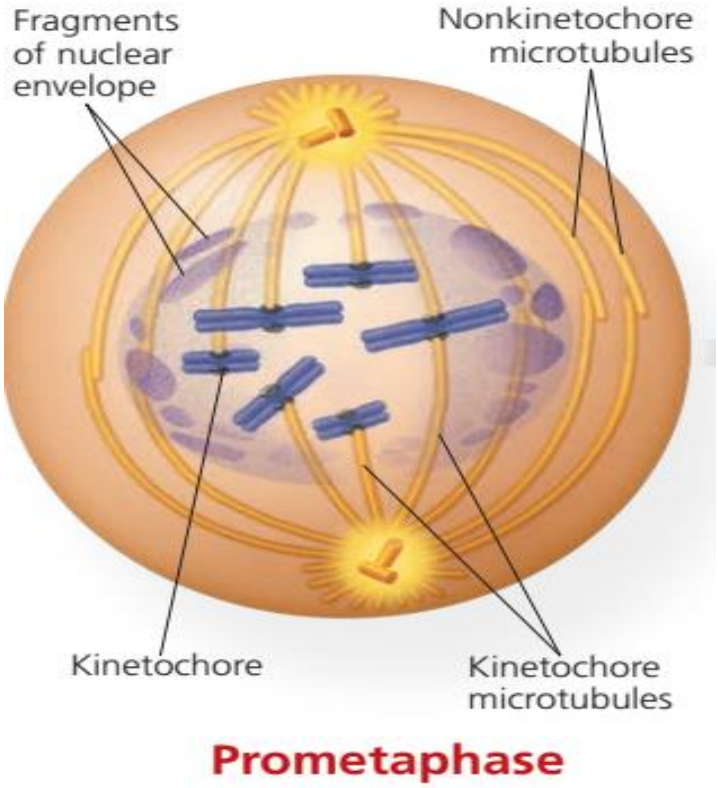
- A nuclear envelope encloses the nucleus
- The nucleus contains one or more nucleoli
- Two centrosomes have formed by duplication of a single centrosome, each centrosome contains two centrioles
- Chromosomes can not be seen individually because they have not yet condensed

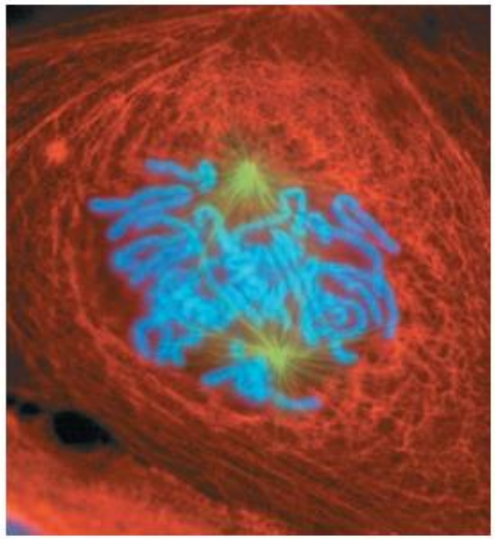


Prophase

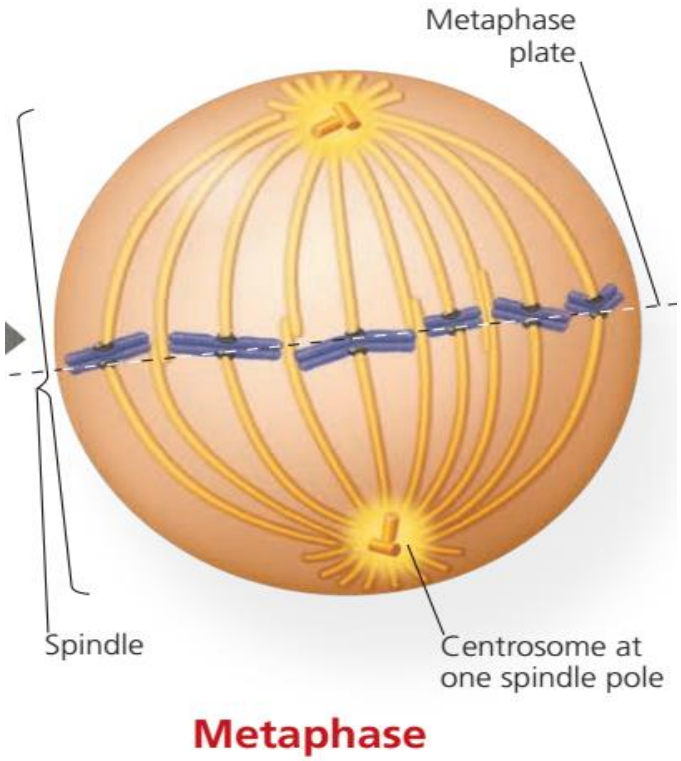


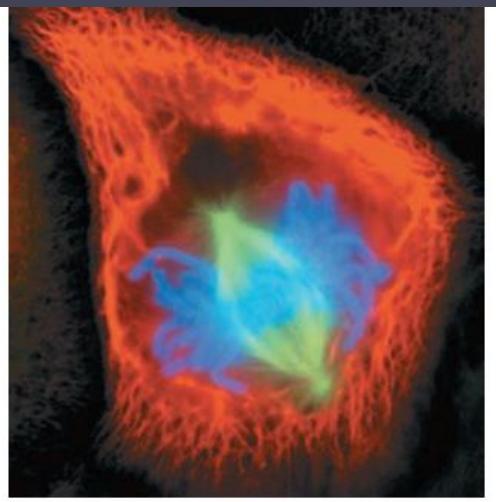
- The chromatin fibers become more tightly coiled
- Each duplicated chromosome appears as two identical sister chromatids joined at their centromere
- The nucleolus disappears
- The mitotic spindle begins to form. The radical arrays of shorter microtubules (spindle fiber) that extend from the centrosomes are called asters (stars)
- The centrosomes move away from each other



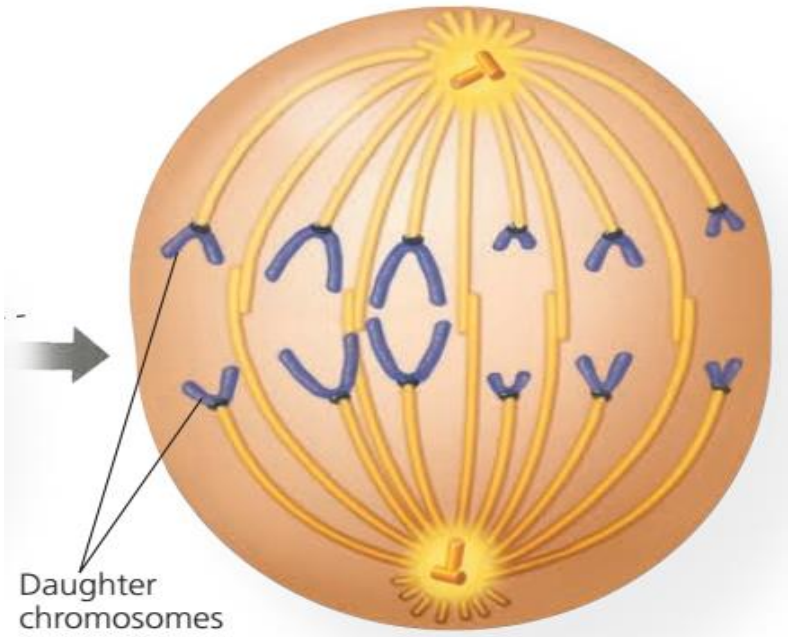


- The nuclear envelope fragments -> the spindle fiber can invade the nuclear area
- The chromosomes become more condensed
- A kinetochore, formed at the centromere of each chromatid
- Microtubules attach to the kinetochore-kinetochore microtubules
- Non-kinetochore microtubules interact with those from the opposite pole of the spindle lengthening the cell

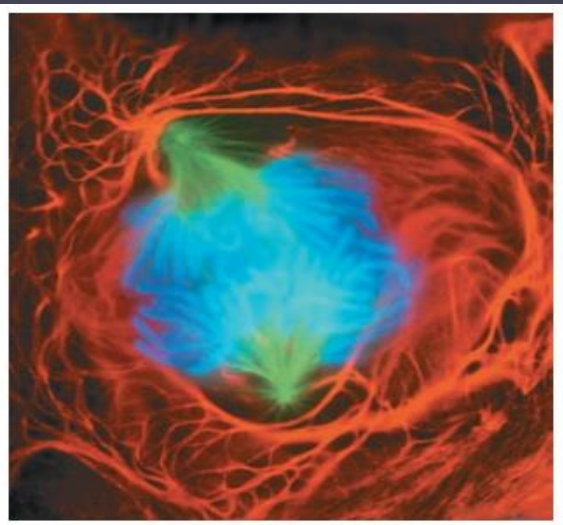




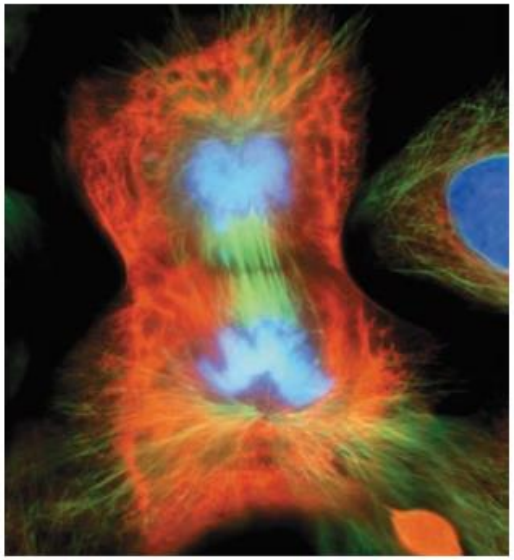
- The centrosomes are now at opposite poles of the cell
- The chromosomes have all arrived at the metaphase plate. The chromosome' centromeres lie at the metaphase plate
- For each chromosome, the kinetochores of the sister chromatids are attached to the kinetochore microtubules coming from opposite poles



Anaphase

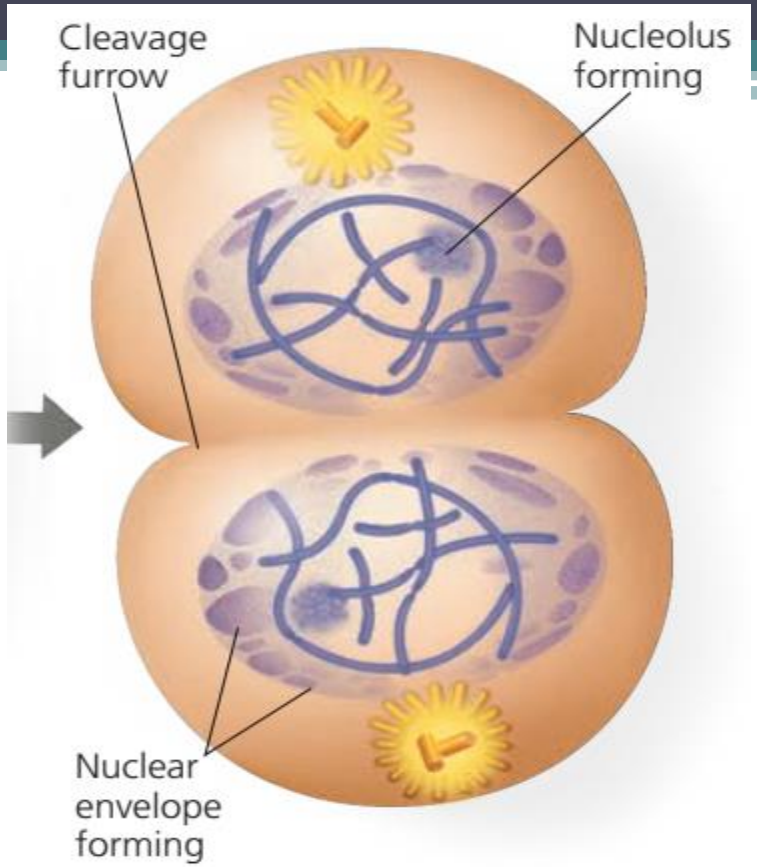


- The shortest stage of mitosis – lasting only a few minutes
- Two sister chromatids of each pair part suddenly. Two new daughter chromosomes moving toward opposite poles of the cells



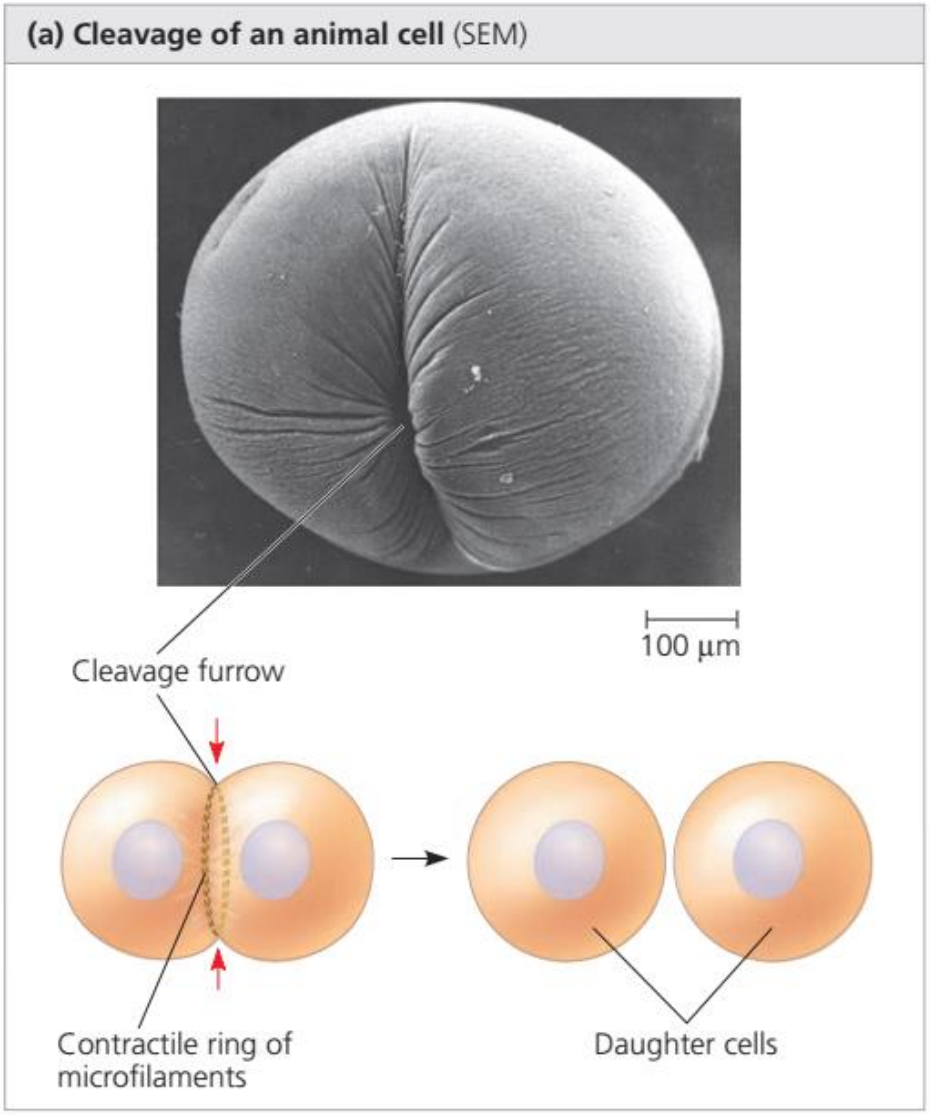
Telophase

- Two daughter nuclei form in the cell. Nuclear envelopes arise from the fragments of the parent cell's nuclear envelope and other portions of the endomembrane system.
- Nucleoli reappear.
- The chromosomes become less condensed.
- Any remaining spindle microtubules are depolymerized.
- Mitosis, the division of one nucleus into two genetically identical nuclei, is now complete.

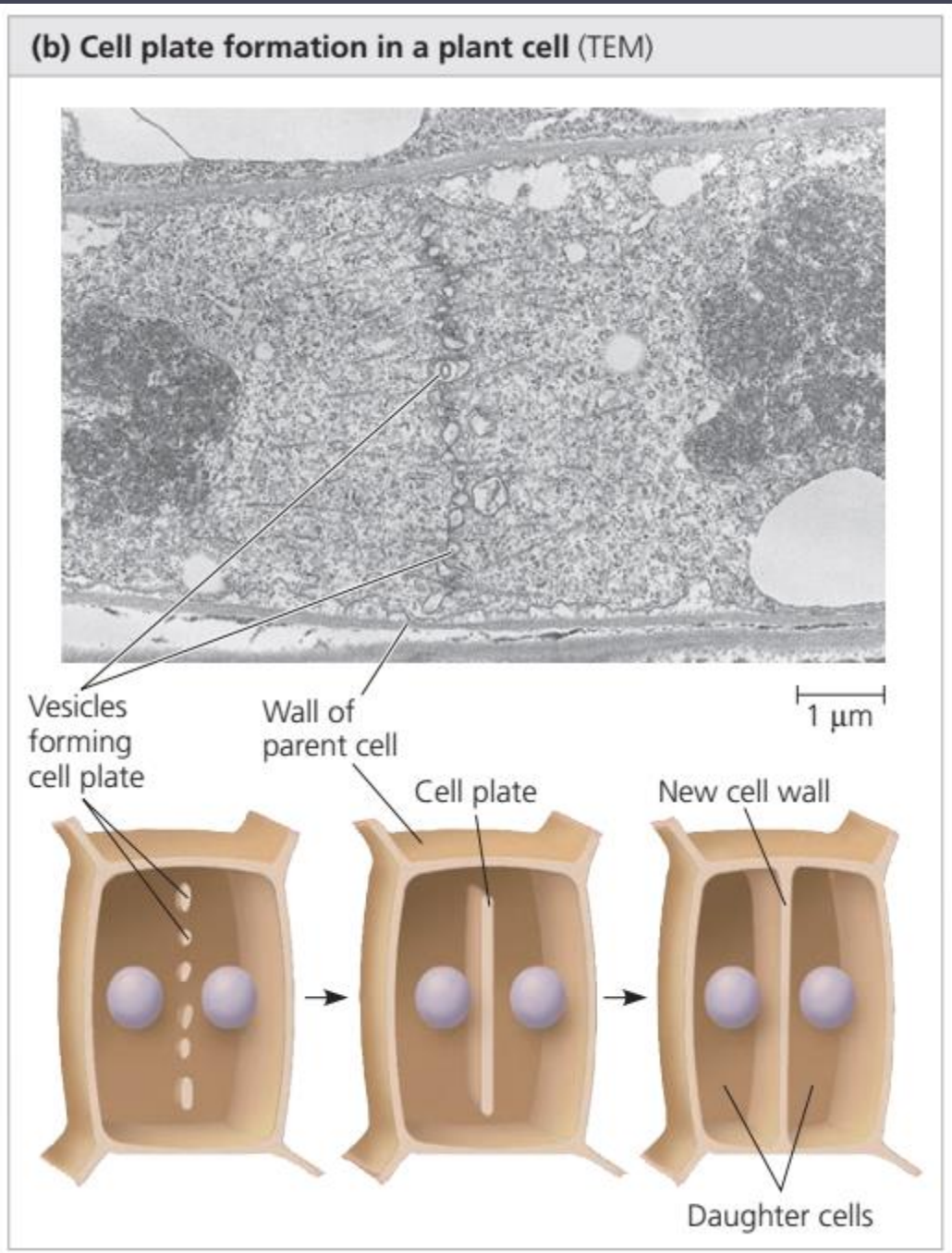


Cytokinesis

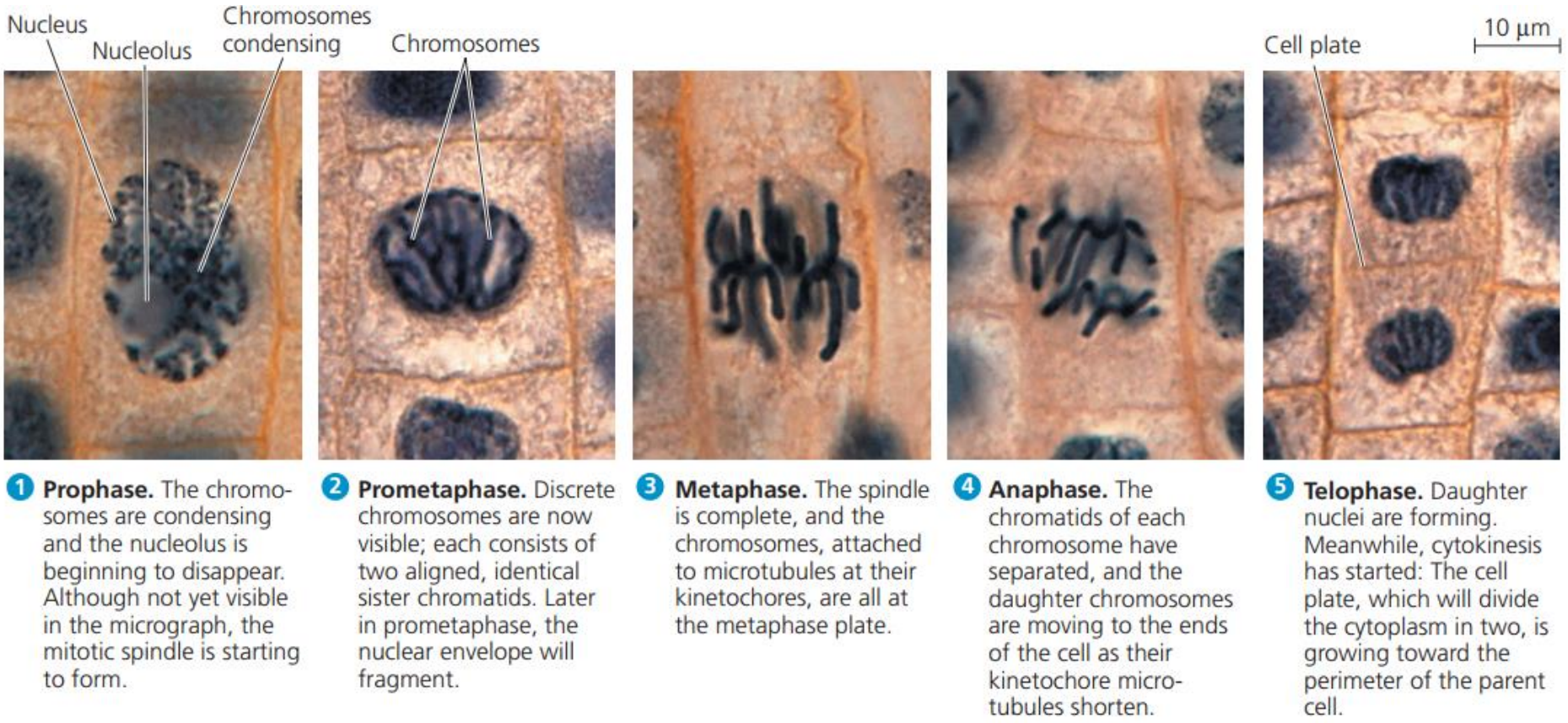
- The division of the cytoplasm is usually well under way by late telophase, so the two daughter cells appear shortly after the end of mitosis.
- In animal cells, cytokinesis involves the formation of a cleavage furrow, which pinches the cell in two.



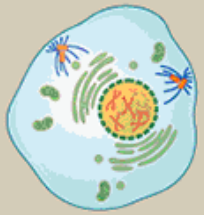
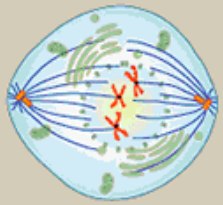
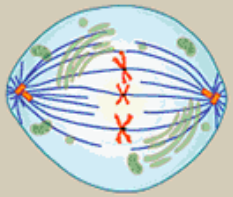
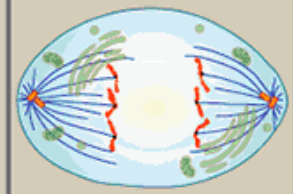
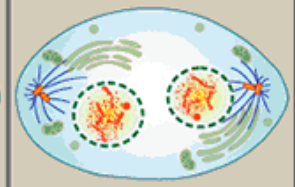
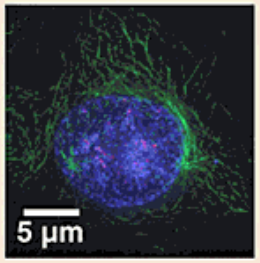
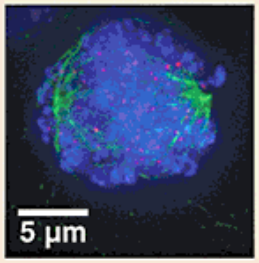
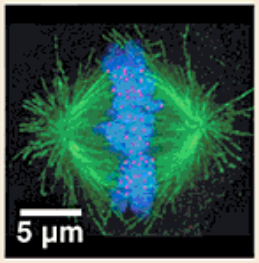
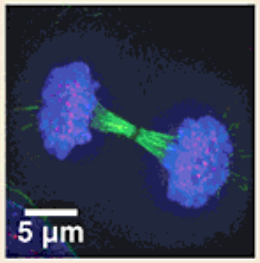
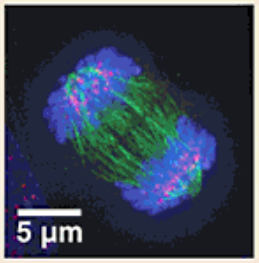
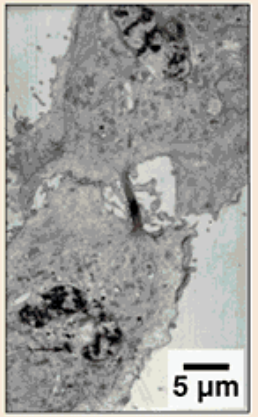
In animal cells, cytokinesis occurs by a process known as **cleavage**. The first sign of cleavage is the appearance of a **cleavage furrow**, a shallow groove in the cell surface near the old metaphase plate



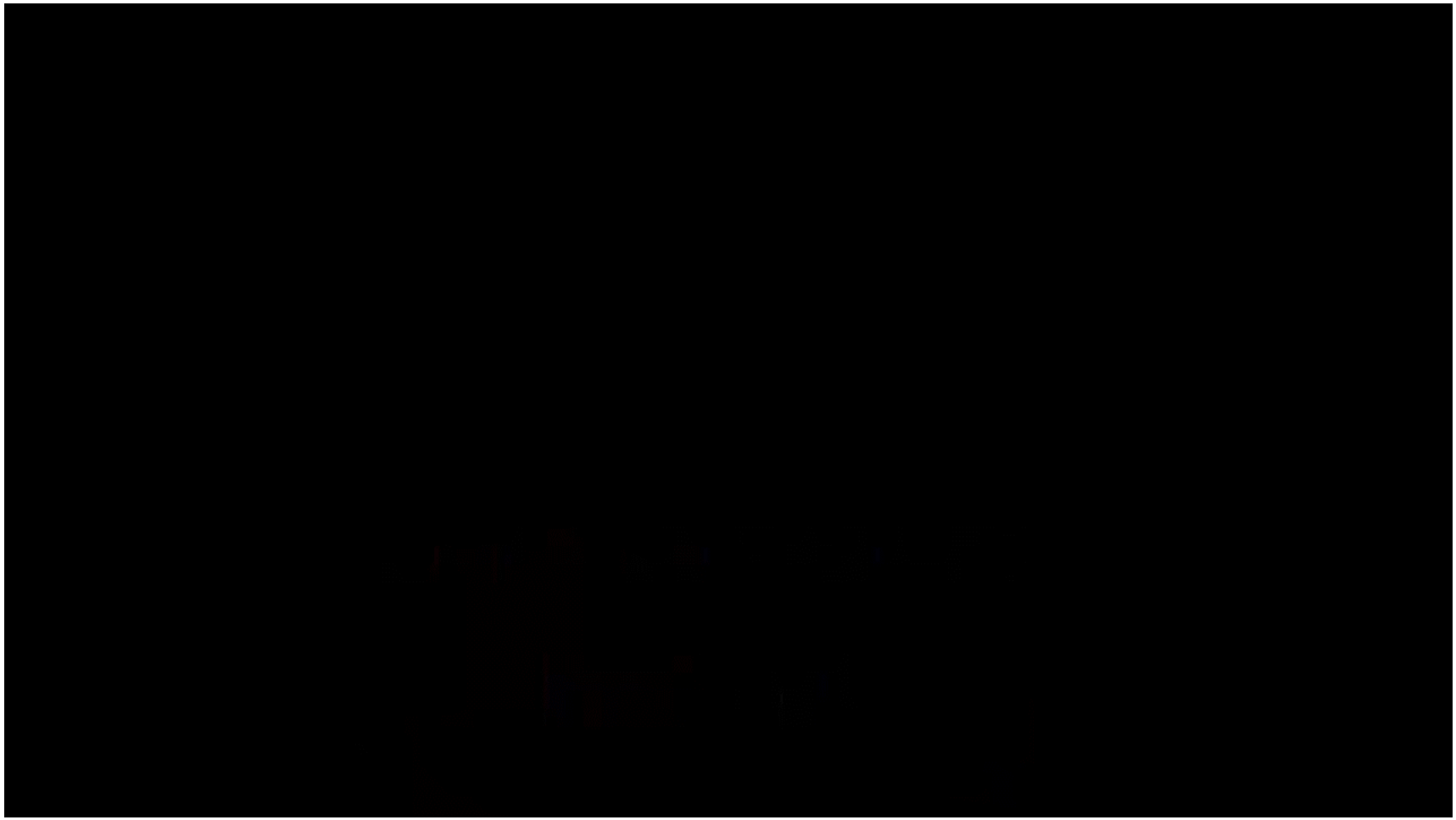
During telophase, vesicles derived from the Golgi apparatus move along microtubules to the middle of the cell, where they coalesce, producing a **cell plate** . Cell wall materials carried in the vesicles collect inside the cell plate as it grows



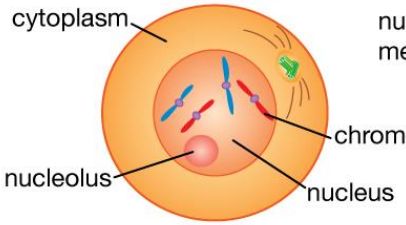
Mitosis in a plant cell. These light micrographs show mitosis in cells of an onion root.

Prophase	Prometaphase	Metaphase	Anaphase	Telophase	Cytokinesis
					
• Chromosomes condense and become visible • Spindle fibers emerge from the centrosomes • Nuclear envelope breaks down • Centrosomes move toward opposite poles	• Chromosomes continue to condense • Kinetochores appear at the centromeres • Mitotic spindle microtubules attach to kinetochores	• Chromosomes are lined up at the metaphase plate • Each sister chromatid is attached to a spindle fiber originating from opposite poles	• Centromeres split in two • Sister chromatids (now called chromosomes) are pulled toward opposite poles • Certain spindle fibers begin to elongate the cell	• Chromosomes arrive at opposite poles and begin to decondense • Nuclear envelope material surrounds each set of chromosomes • The mitotic spindle breaks down • Spindle fibers continue to push poles apart	• Animal cells: a cleavage furrow separates the daughter cells • Plant cells: a cell plate, the precursor to a new cell wall, separates the daughter cells
					

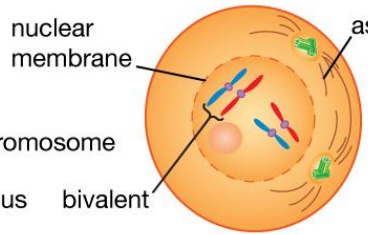
MITOSIS



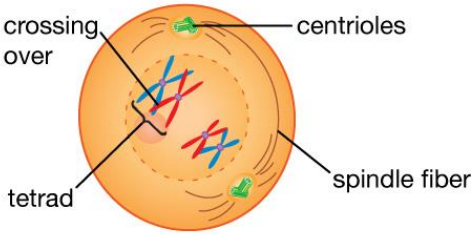
Meiosis, or sex cell division



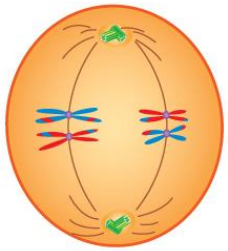
At the onset of meiosis, DNA strands thicken into chromosomes. Homologous, or like, chromosomes begin to approach each other.



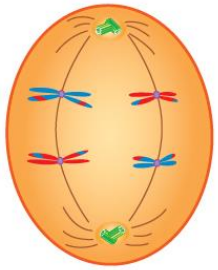
Homologous chromosomes pair to form bivalents. The centrioles divide and move to opposite poles of the cell.



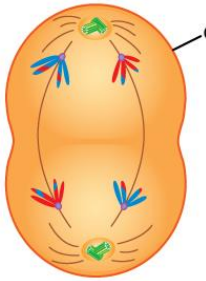
The bivalents duplicate to form tetrads, or four-chromatid groups. The nuclear membrane disintegrates. Crossing over (recombination) occurs.



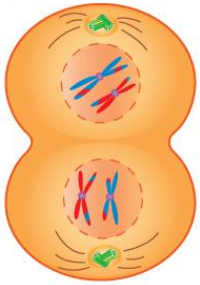
In metaphase I, the tetrads, attached to spindle fibers at their centromeres, line up at mid-cell.



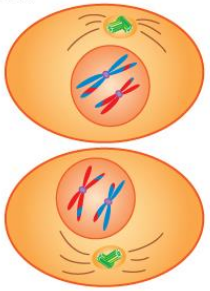
In early anaphase I, the tetrads separate, and the paired chromatids move along the spindle to their respective centrioles.



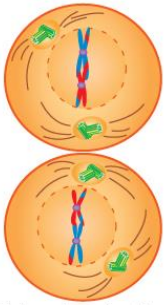
In late anaphase I, the chromatids have almost reached the spindle poles. The cell membrane begins to constrict.



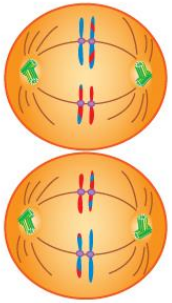
In telophase I, nuclear membranes enclose the separated chromatids. The cell membrane completes its constriction.



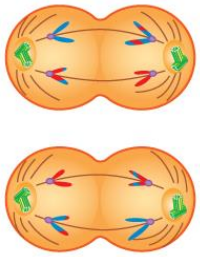
The first meiotic division ends. There are now two cells, each with the same number of chromatids as the parent cell.



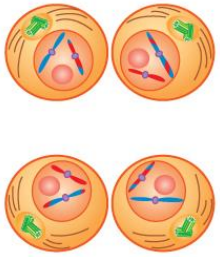
Prophase II begins. In the second meiotic division, homologous chromatids do not duplicate but merely separate.



In metaphase II, the chromatids line up at mid-cell. The centrioles and asters are at the poles. A spindle has formed.



In anaphase II, the now-separated chromatids approach their respective poles. The cell membrane begins to constrict.



Telophase II has been completed. There are now four cells, each with half the number of chromosomes of the parent cell.

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